■ FloodSense India - Flood Risk Prediction Report

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Executive Methodology Summary

Flood susceptibility modeling was conducted using an ensemble of machine learning algorithms including xgboost, random_forest, lightgbm, decision_tree, logistic_regression. The feature space comprised 21 spatial conditioning factors (Latitude, Longitude, Rainfall (mm), Temperature (°C), Humidity (%), River Discharge (m³/s), Water Level (m), Elevation (m), Population Density, Land Cover_Agricultural, Land Cover_Desert, Land Cover_Forest, Land Cover_Urban, Land Cover_Water Body, Soil Type_Clay, Soil Type_Loam, Soil Type_Peat, Soil Type_Sandy, Soil Type_Silt, Infrastructure, Historical Floods), standardized to a common grid in EPSG:4326 projection. Ensemble predictions were generated using both pixel-wise probability soft voting and a meta-learner stacking strategy. Classification into five susceptibility levels was executed using Equal Interval and Jenks Natural Breaks optimization. Spatial block cross-validation and feature importance analysis via SHAP and permutation importance were evaluated to prevent spatial data leakage and assess feature attributions.

Scientific Limitations & Disclaimers

1. Susceptibility models represent long-term spatial likelihood based on static terrain and historical factors, not real-time hydrological flood forecasts.
2. SHAP feature contributions quantify statistical associations and must not be interpreted as direct causal drivers.
3. Field verification by local water resource authorities is recommended prior to emergency policy deployment.